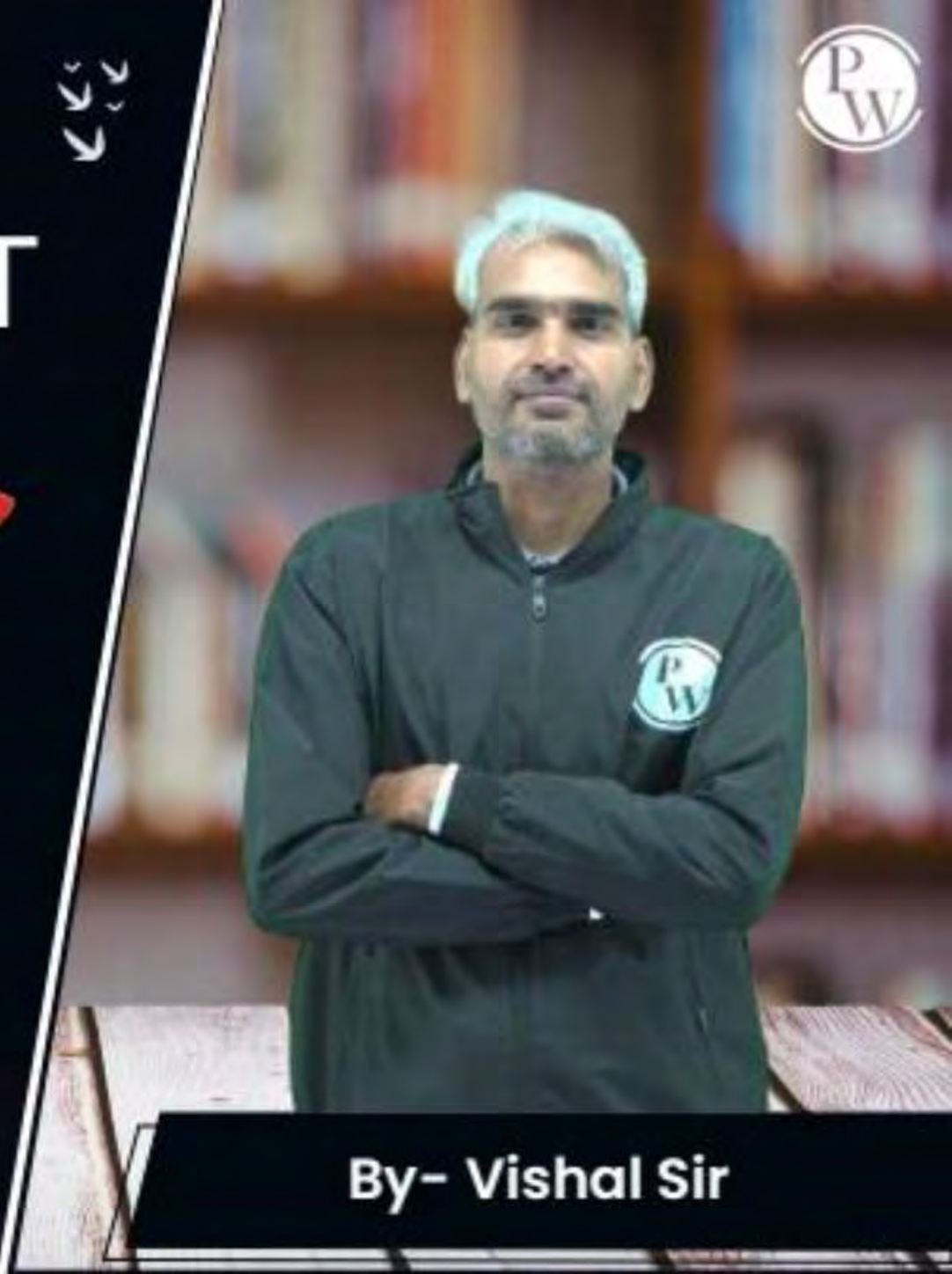
# Computer Science & IT



## Database Management System

## Entity Relationship Model & Integrity constraints

Lecture No. 02



By- Vishal Sir

# Recap of Previous Lecture



**Topic** ER model & ER diagram

**Topic** Relational model & Integrity constraints

Slide

# Topics to be Covered

| Topic | ER model to RDBMS table |
| --- | --- |

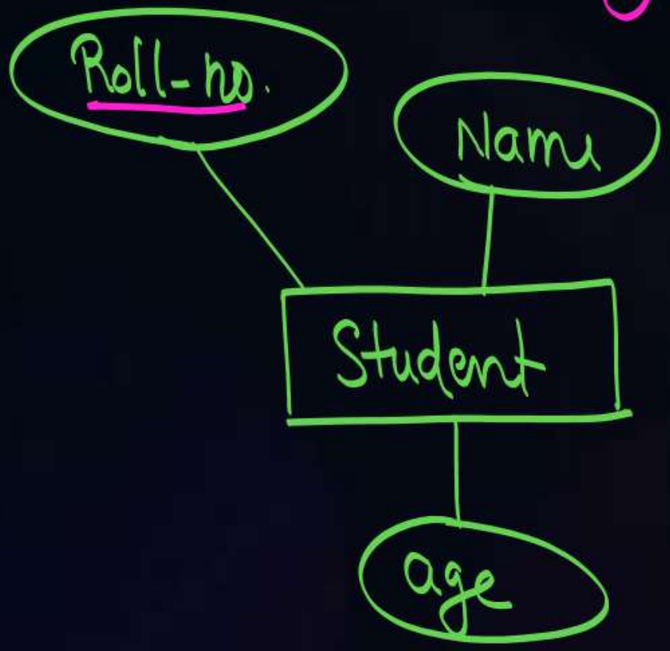


Slide

# Topic : ER model to RDBMS table

We have ER model ⟶ we need ⟶ RDBMS table

(i) Entity set → will be converted into relation (table)




ER Model

all attributes in ER diagram ⟹ are simple & single valued attribute

Student

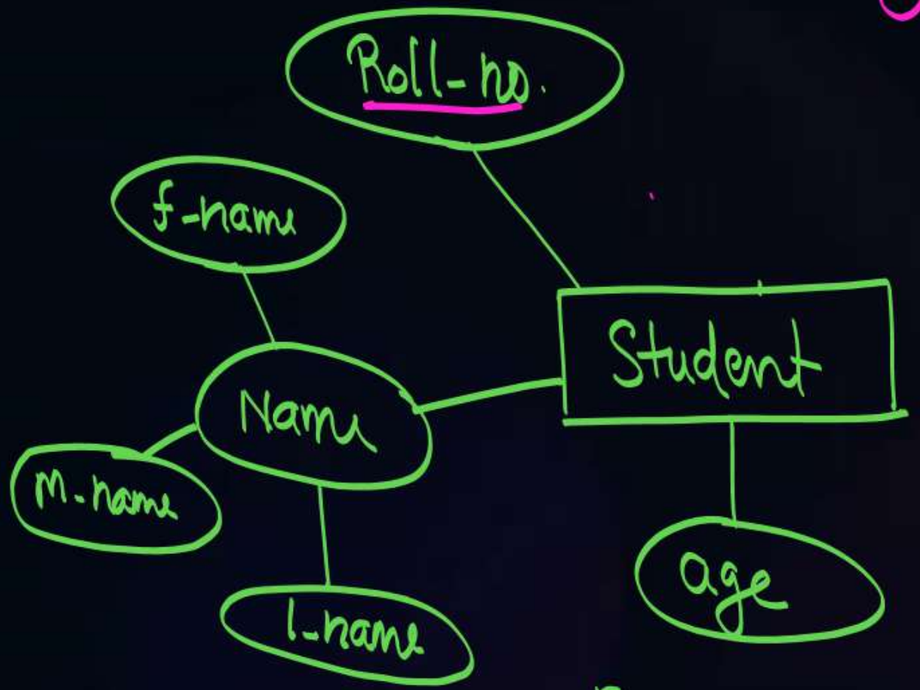
| Roll-no | Name | age |
|---|---|---|
| | Ram | |
| | Mohan | |
| | Sita | |

Relational Model

# Topic : ER model to RDBMS table

We have ER model ⟶ we need ⟶ RDBMS table

(ii) Entity set → will be converted into relation (table)




ER Model

all attributes in ER diagram ⟹ Except 'Name' are simple & single valued attribute

Student

| Roll-no | f-name | m-name | l-name | age |
|---|---|---|---|---|
| | we will store only simple components of composite attribute in Relational Model | | | |

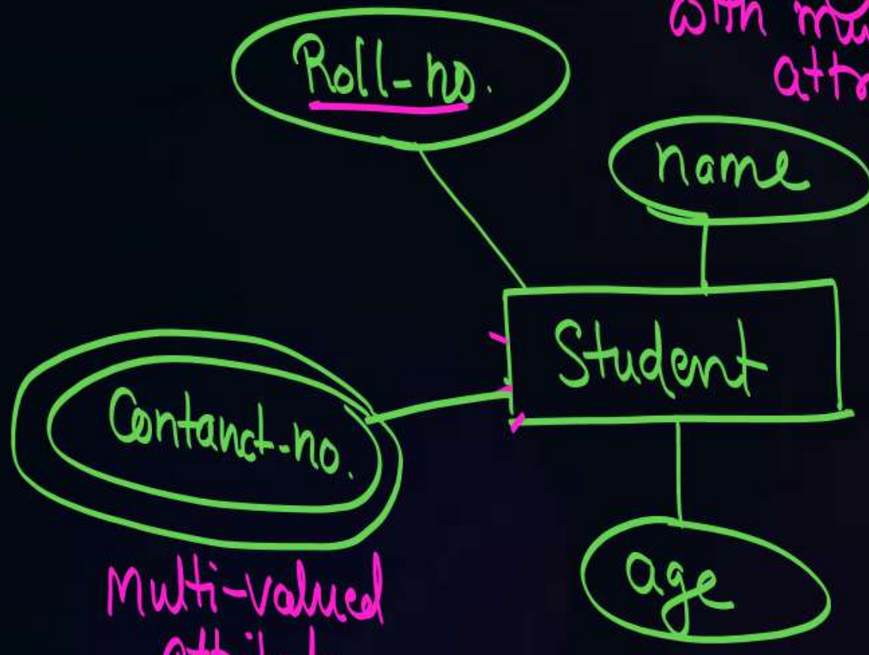
Relational Model



# Topic : ER model to RDBMS table

We have ER model ⟶ we need ⟶ RDBMS table

(iii) Entity set → will be converted into relation (table)
with multivalued attribute




Multi-valued attribute

ER Model

two

Student

all attributes in ER diagram ⟹ Except 'Contact-no' are Simple & Single Valued attribute

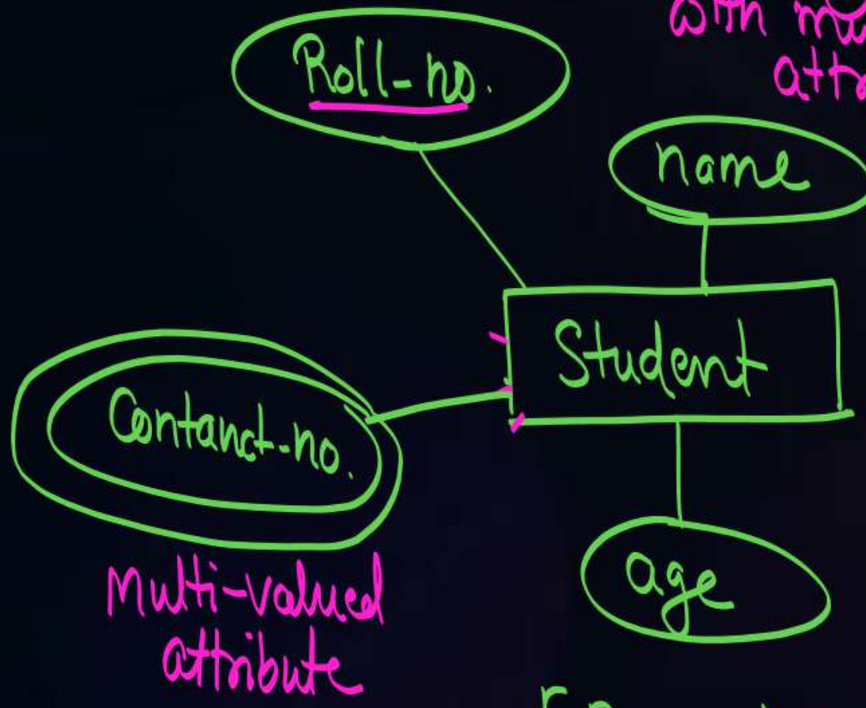
{ Note: Seperate table will be required in relational model w.r.t. multivalued attribute }

# Topic : ER model to RDBMS table

We have ER model ⟶ we need ⟶ RDBMS table

(iii) Entity set → will be converted into relation (table)

with multivalued attribute




Multi-valued attribute

ER Model

all attributes in ER diagram

Except 'Contact-no' are Simple & Single Valued attribute

Two table

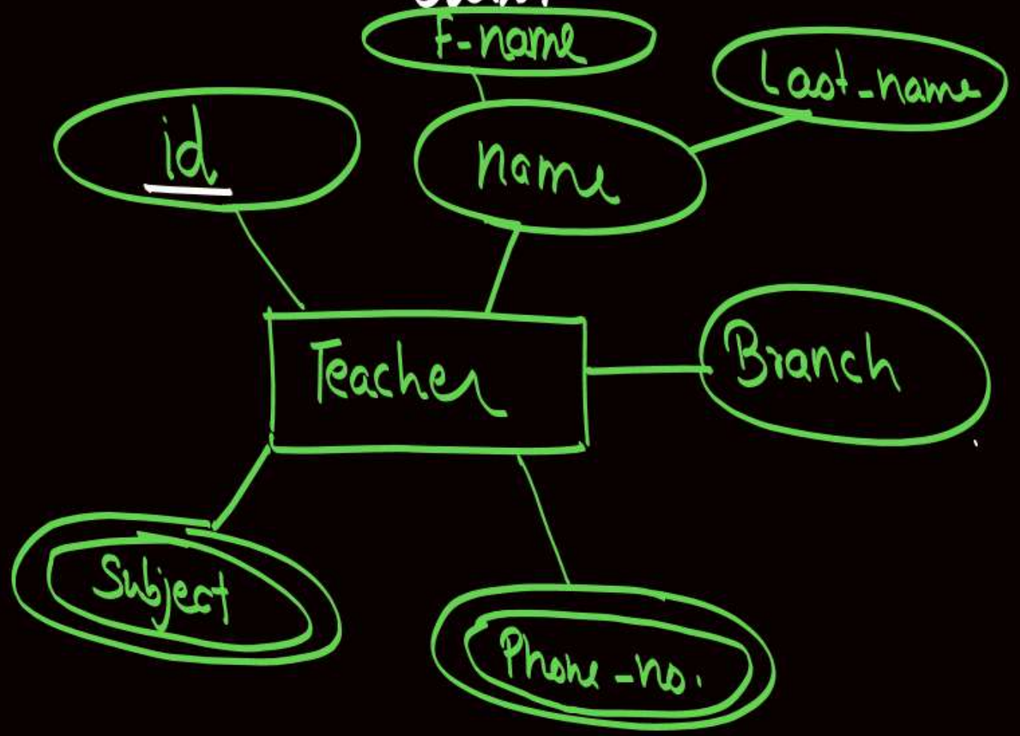
Student

| Roll-no | name | age |
|---|---|---|
| | | |

Contact-info

| Roll-no. | Contact-no. |
|---|---|
| | |

Multi-valued Attributes exist → then → two tables required




⇒

Teacher

| id | f_name | last_name | Branch |
|---|---|---|---|
| | | | |

1. Key attribute + all simple & Single Valued attributes + all simple Components of Composite attributes

2. Key attributes + all multivalued attributes of entity

Teacher_two

| id | Subject | Phone_no |
|---|---|---|
| | | |

* There is no representation of Composite attributes and multi-valued attributes in relational model.



# Topic : ER model to RDBMS table

## ① One-to-one Binary relation (1 : 1)

① 

{ 2 tables are required }

② 

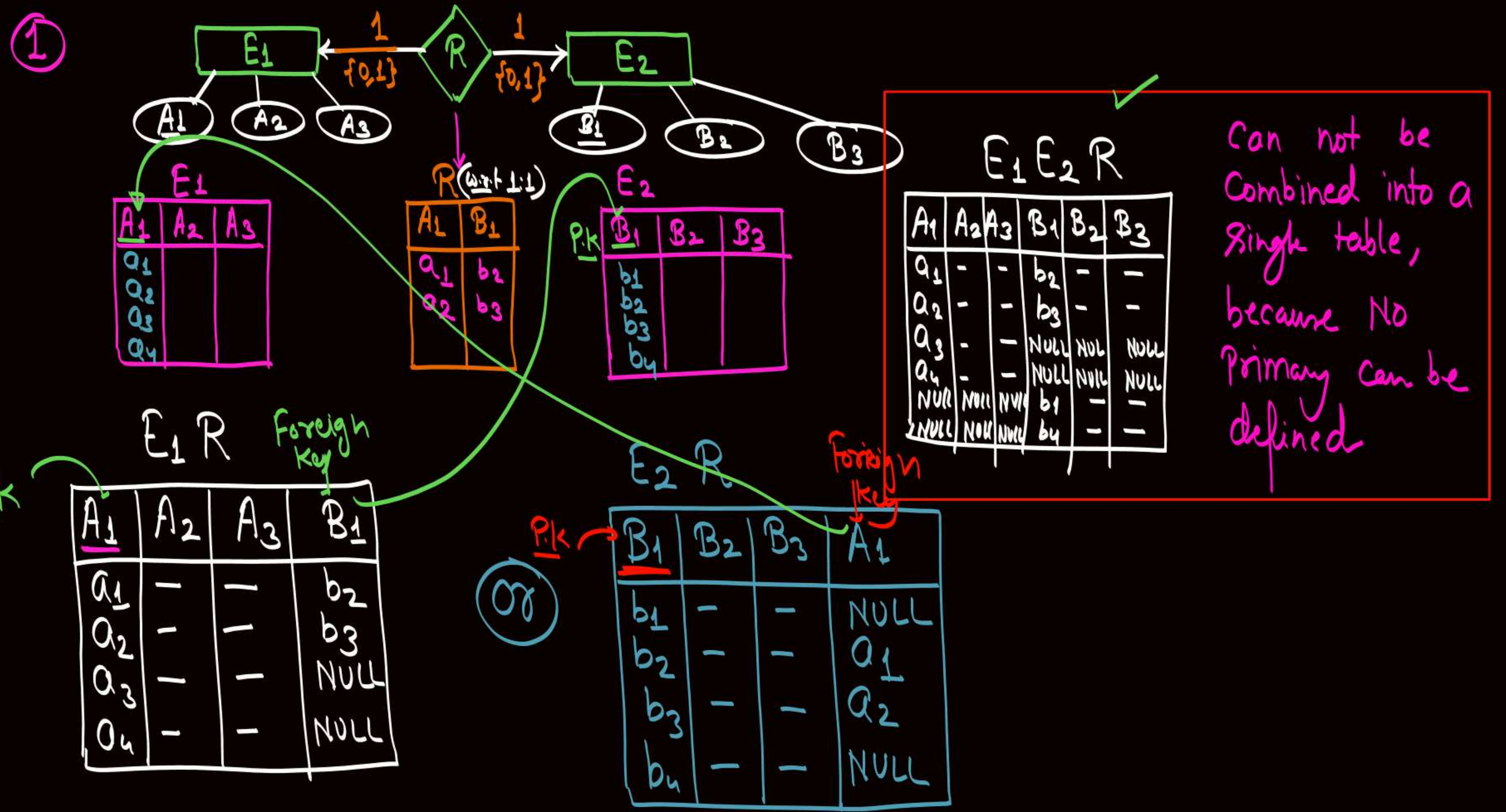
{ only one table }

③ 

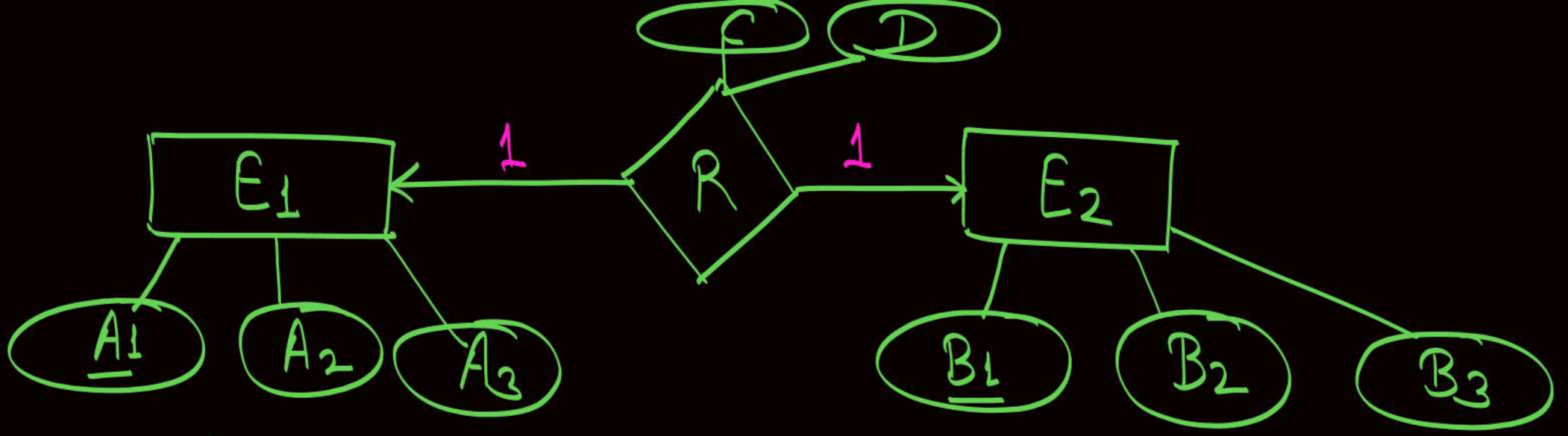
{ only one table }

④ 

{ only one table }


1
E1
1
{0,1}
R
1
{0,1}
E2
A1
A2
A3
B1
B2
B3
E1
A1 A2 A3
a1
a2
a3
a4
R (wrt 1:1)
A1 B1
a1 b2
a2 b3
E2
P.K
B1 B2 B3
b1
b2
b3
b4
E1 E2 R
A1 A2 A3 B1 B2 B3
a1 - - b2 - -
a2 - - b3 - -
a3 - - NULL NUL NULL
a4 - - NULL NUL NULL
NUL NULL NUL b1 - -
NULL NUL NUL b4 - -
Can not be Combined into a Single table, because No Primary can be defined
E1 R
Foreign Key
P.K
A1 A2 A3 B1
a1 - - b2
a2 - - b3
a3 - - NULL
a4 - - NULL
E2 R
Foreign Key
P.K
B1 B2 B3 A1
b1 - - NULL
b2 - - a1
b3 - - a2
b4 - - NULL
or

(1)




Minimum two tables are required

P.K F.K

(1) $E_1R(\underline{A_1}, A_2, A_3, B_1, C, D)$

&

$E_2(\underline{B_1}, B_2, B_3)$

OR

(2)

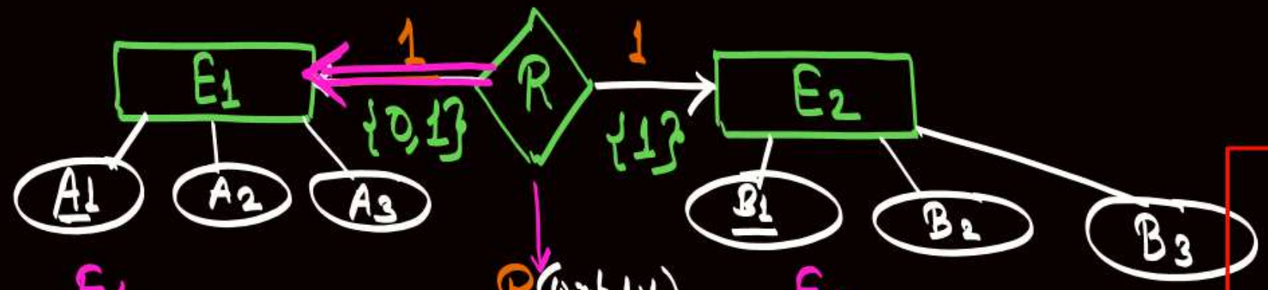
P.K. F.K

$E_2R(\underline{B_1}, B_2, B_3, A_1, C, D)$

&

$E_1(\underline{A_1}, A_2, A_3)$

2




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |

R (w.r.t 1:1)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a2 | b3 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |
| b4 | | |

P.K ✓

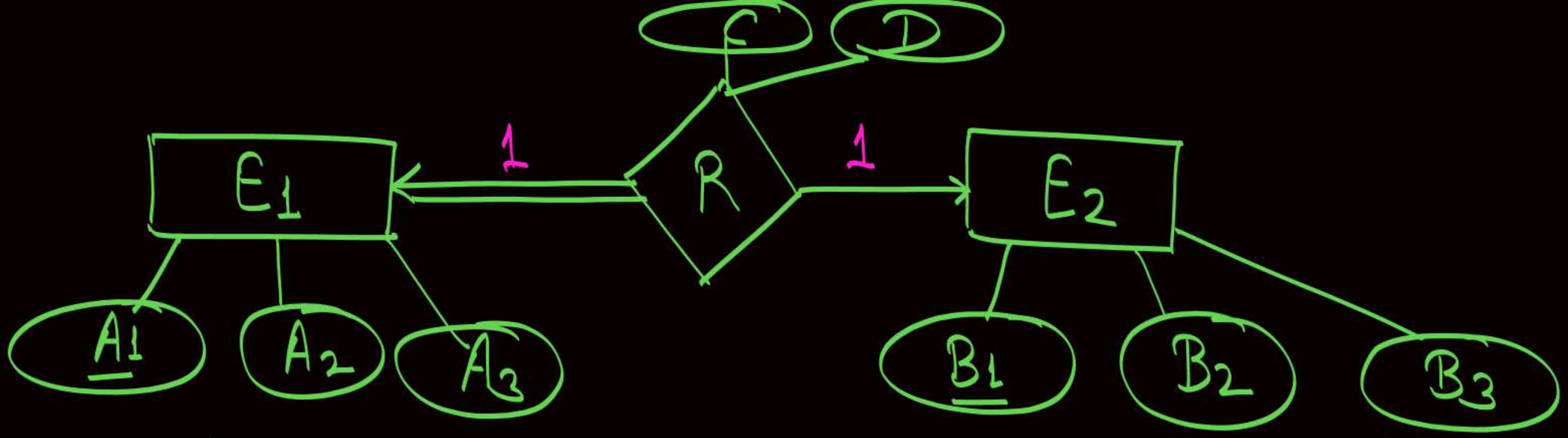
E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a2 | - | - | b3 | - | - |
| NUL | NUL | NUL | b1 | - | - |
| NL | NL | NL | b4 | - | - |

'B1' will become Primary key

Can be Combined into a Single table, if any of the entity w.r.t. 1:1 is totally Participating

(2)

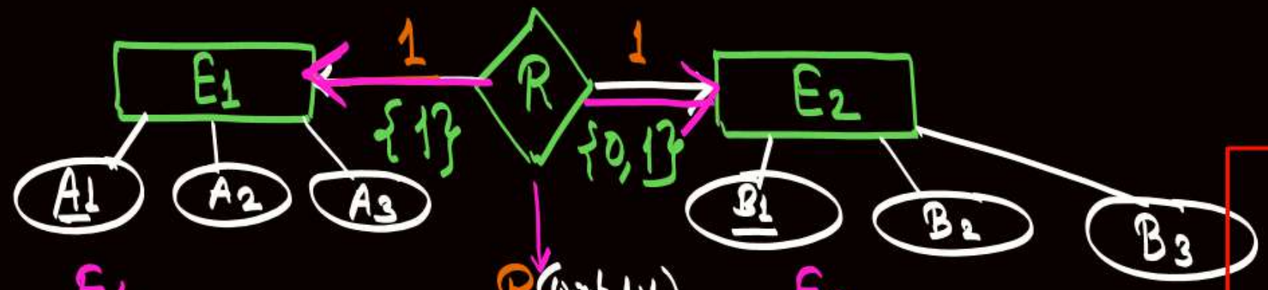



Only one table will be required

$E_1E_2R(\underline{B_1}, B_2, B_3, A_1, A_2, A_3, C, D)$

Primary key of the relation with partial participation will become the primary key of merged table.

(3)




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | – | – |
| a2 | – | – |
| a3 | – | – |
| a4 | – | – |

R (w.r.t 1:1)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a2 | b3 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b2 | – | – |
| b3 | – | – |

P.K

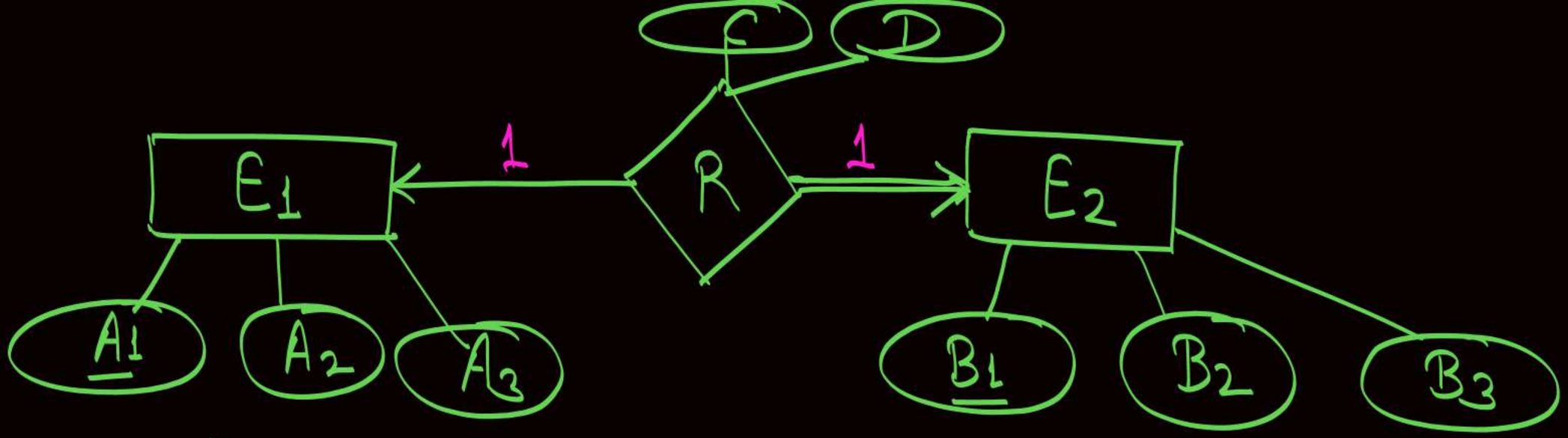
E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a2 | - | - | b3 | - | - |
| a3 | - | - | NL | NL | NL |
| a4 | - | - | NL | NL | NL |

'A1' will become Primary key

Can be Combined into a Single table, if any of the entity w.r.t. 1:1. is totally Participating

(2)

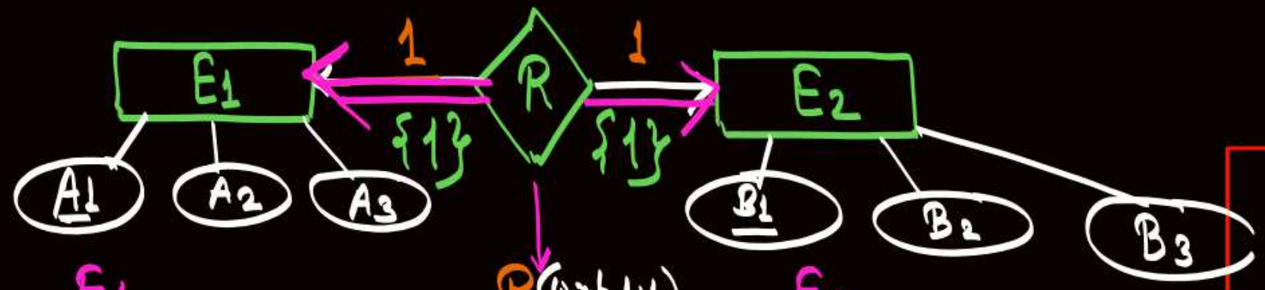



Only one table will be required

$E_1E_2R(\underline{A_1}, A_2, A_3, B_1, B_2, B_3, C, D)$

Primary key of the relation with partial participation will become the primary key of merged table.

4




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | - | - |
| a2 | - | - |

R (w.r.t 1:1)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a2 | b3 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b2 | - | - |
| b3 | - | - |

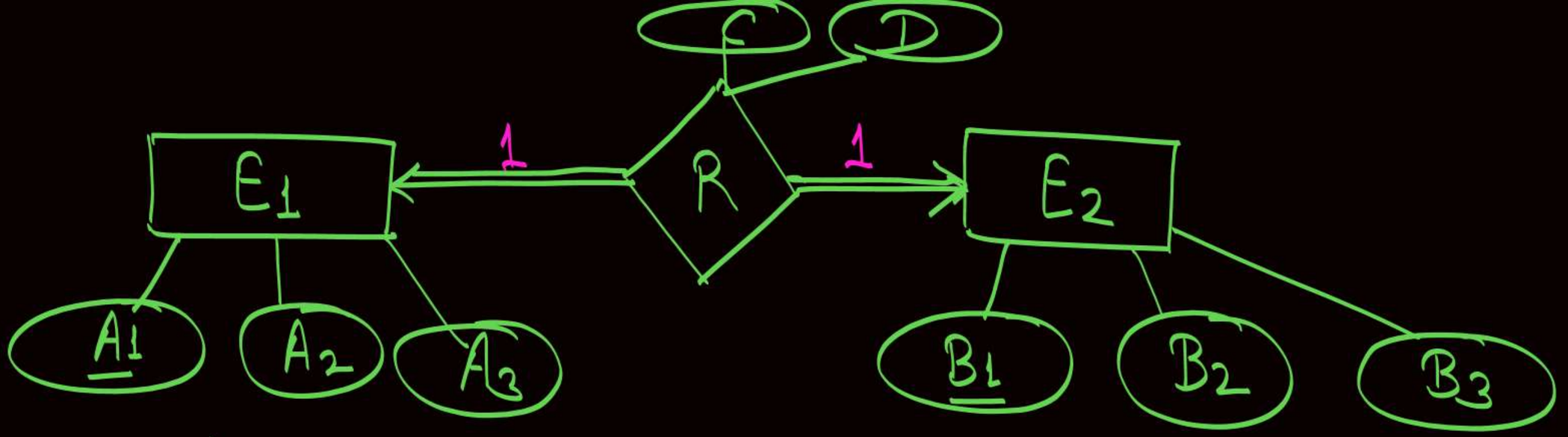
E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | (a0) | b2 | - | - |
| a2 | - | - | b3 | - | - |

Can be Combined into a Single table, if any of the entity w.r.t. 1:1 is totally Participating

We can choose A1 or B1 as Primary key

(4)




Only one table will be required

$E_1E_2R(\underline{A_1}, A_2, A_3, B_1, B_2, B_3, C, D)$

Primary key can be

$A_1$ or $B_1$

{ Primary key of any of the entity set can be Primary key of Merged table.



# Topic : ER model to RDBMS table

② One-to-Many Binary relation (1 : N)




{ Minimum two tables }




{ only one table is required }

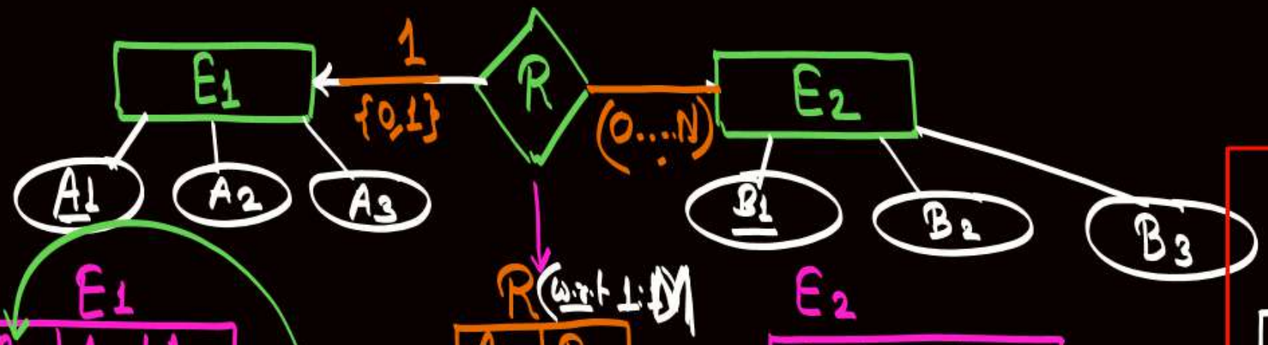



{ Minimum two tables are required }




{ only one table is required }

(1)




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |
| a3 | | |
| a4 | | |

R (with 1:N)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b4 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |
| b4 | | |

✓

E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b4 | - | - |
| a3 | - | - | N | N | N |
| a4 | - | - | N | N | N |
| N | N | N | b1 | N | N |

Can not be Combined into a Single table, because No Primary can be defined

Primary key can not be defined

E1 R

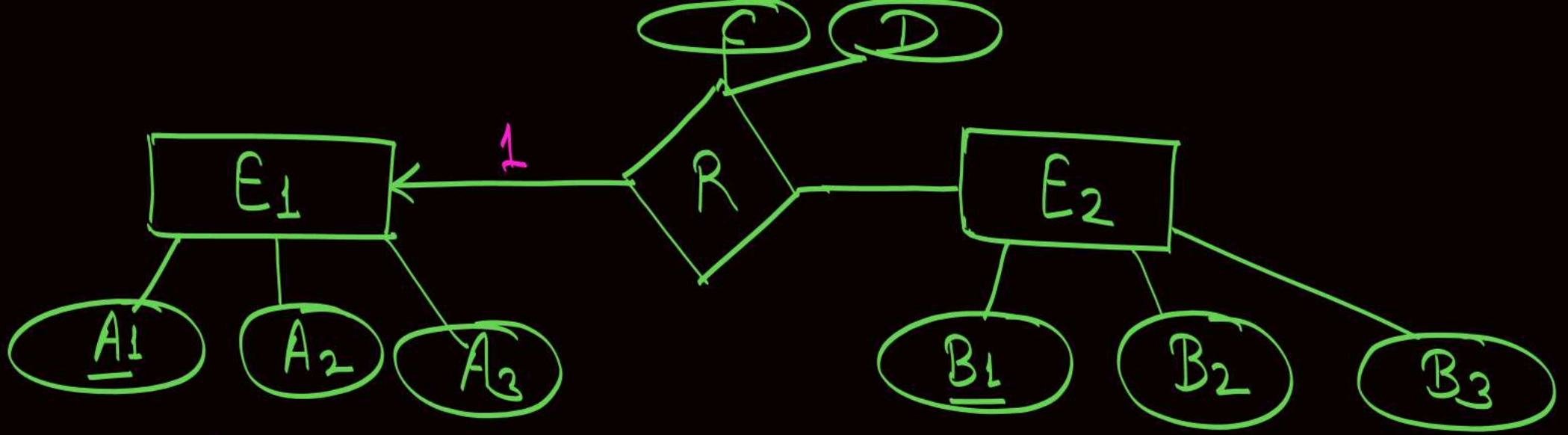
| A1 | A2 | A3 | B1 |
|---|---|---|---|
| a1 | - | - | b2 |
| a1 | - | - | b3 |
| a2 | - | - | b4 |
| a3 | - | - | NULL |
| a4 | - | - | NULL |

or

E2 R (Foreign key)

| P.K B1 | B2 | B3 | A1 |
|---|---|---|---|
| b1 | - | - | NULL |
| b2 | - | - | a1 |
| b3 | - | - | a1 |
| b4 | - | - | a2 |

(1)




Minimum two tables are required

P.K. F.K

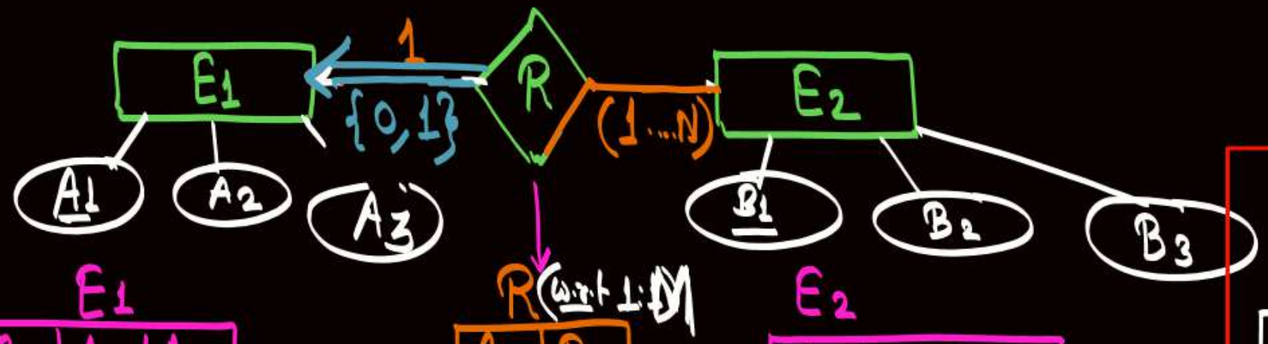
$E_2R(\underline{B_1}, B_2, B_3, A_1, C, D)$

&

$E_1(\underline{A_1}, A_2, A_3)$

} Relationship set will be merged with many side entity set.

(2)




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |

R (w.r.t 1:N)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b4 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |
| b4 | | |

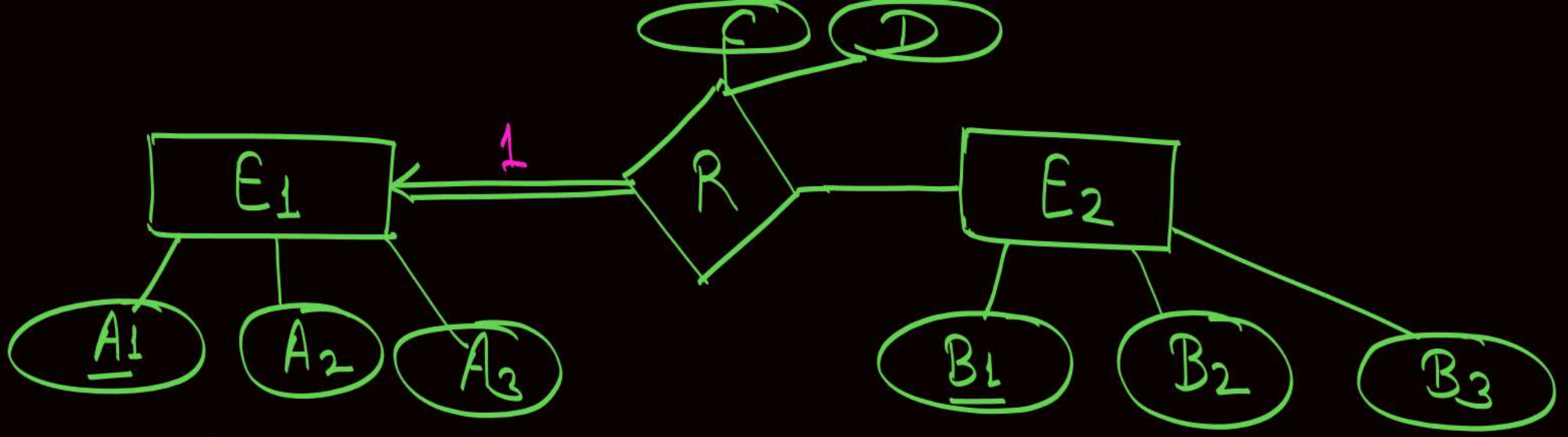
✓

E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b4 | - | - |
| N | N | N | b1 | - | - |

Can be Combined into a Single table, if one side entity totally Participate in the Relation

(2)

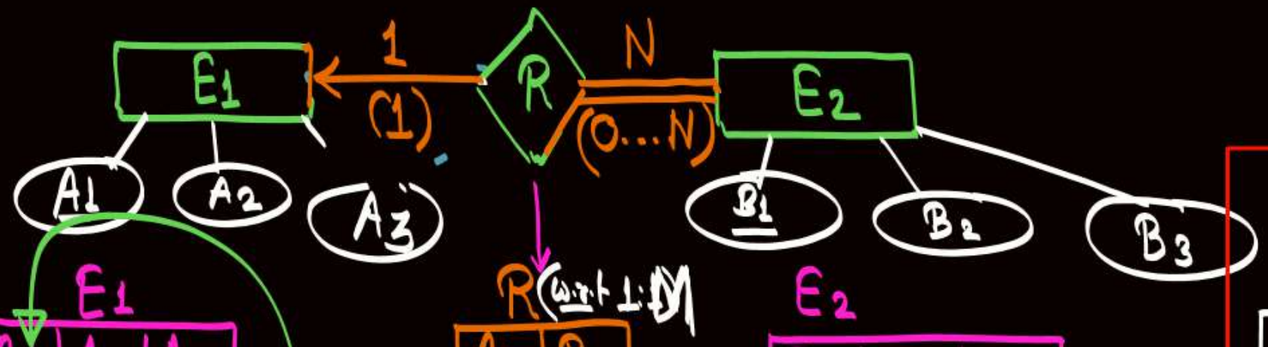



Minimum one tables is required

$E_1 E_2 R (\underline{B_1}, B_2, \underline{B_3}\ A_1, A_2, A_3, C, D\}$

↑ Primary key of many side entity will become primary key of merged relation

(3)




R (with 1:N)

E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |
| a3 | | |

R

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b1 |

E2

P.K

| B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |

✓

E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b1 | - | - |
| a3 | - | - | N | N | N |

Can not be Combined into a Single table, because P.K can not be defined.

Primary key can not be defined

E1 R

| A1 | A2 | A3 | B1 |
|---|---|---|---|
| a1 | | | b2 |
| a1 | | | b3 |
| a2 | | | b1 |
| a3 | | | Null |

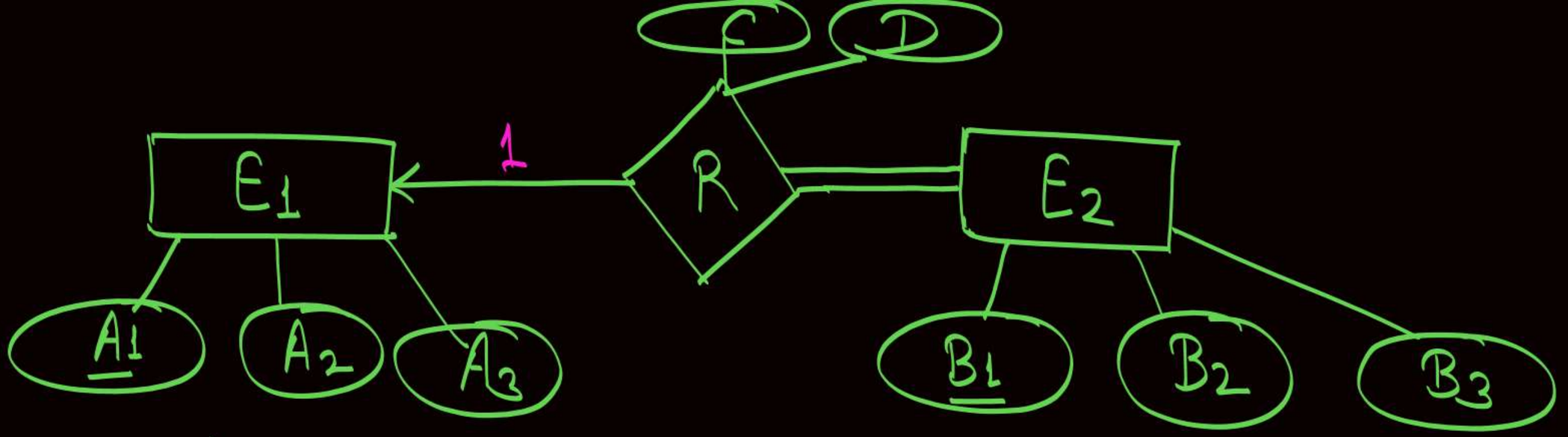
E2 R

F.K

P.K

| B1 | B2 | B3 | A1 |
|---|---|---|---|
| b2 | - | - | a1 |
| b3 | - | - | a1 |
| b1 | - | - | a2 |

(3)




Minimum two tables are required

$E_1(\underline{A_1}, A_2, A_3)$

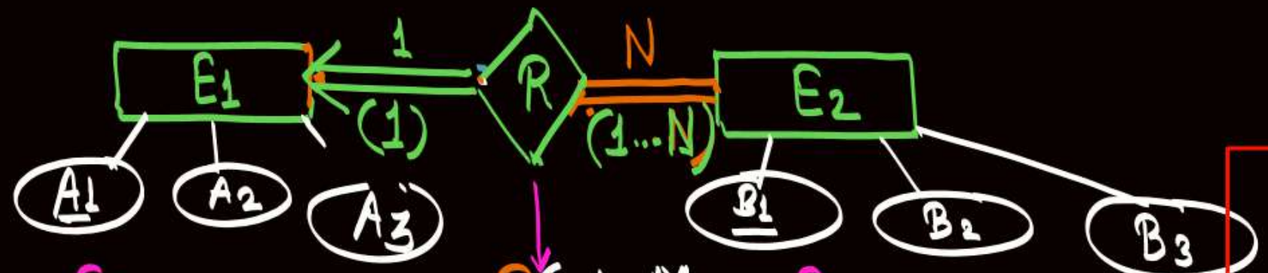
P.K.

$E_2R(\underline{B_1}, B_2, B_3, A_1, C, D)$

P.K. F.K

- Relationship will be merged with Many side entity, and
- Primary key of Many side entity will become P.K. of merged Rel$^n$.

4




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |

R (with 1:N)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b1 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |

✓

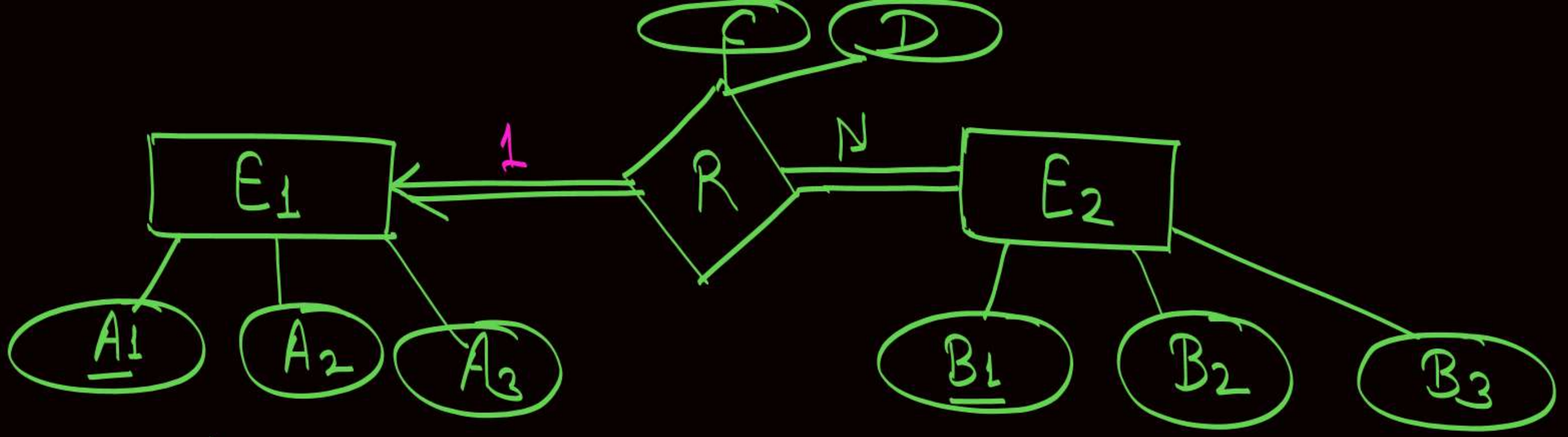
E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b1 | - | - |

Can be Combined into a Single table, if both side entity totally Participate in Rel^n.

Primary key of many side each will become primary key of merged relation.

(4)




Minimum one table is required

$E_1E_2R(\underline{B_1}, B_2, B_3, A_1, A_2, A_3, C, D)$

P.K

Primary key of Many side entity will become P.K. of merged Relⁿ.

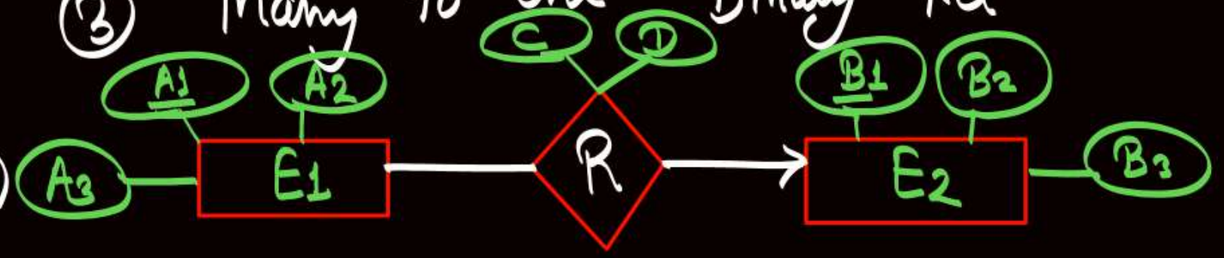


# Topic : ER model to RDBMS table

③ Many-to-one Binary relation { It will be same as One to Many }

## ③ Many to one Binary Rel$^n$.

(i)



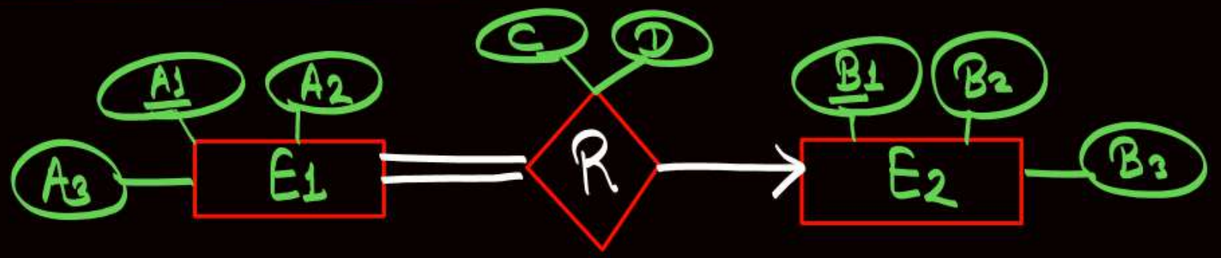

Two tables required

$E_1R(\underline{A_1}, A_2, A_3, \overset{FK}{B_1}, C, D)$

$E_2(\underline{B_1}, B_2, B_3)$

---

(ii)




Two tables required

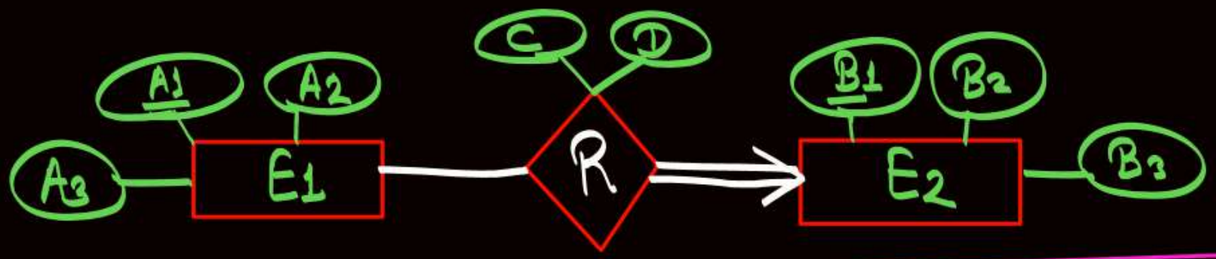
$E_1R(\underline{A_1}, A_2, A_3, \overset{FK}{B_1}, C, D)$

$E_2(\underline{B_1}, B_2, B_3)$

{ If total participation is at many side then it will have no effect in 1:N or ~~N:1~~

---

(iii)



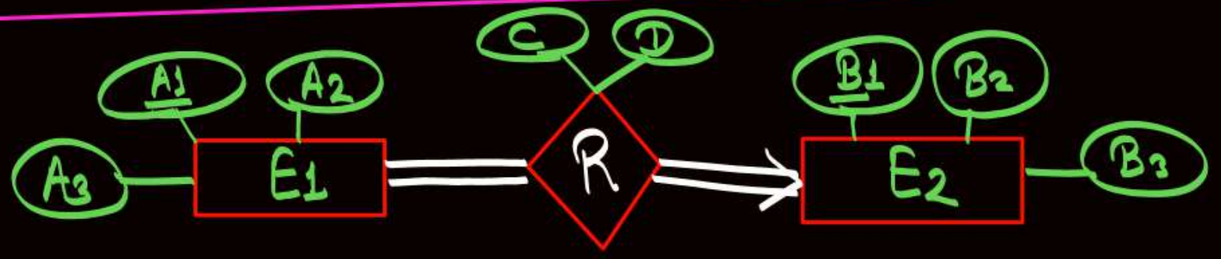

only one table is required

$E_1E_2R(\underline{A_1}, A_2, A_2, B_1, B_2, B_3, C, D)$

P.K will be from many side rel$^n$

---

(iv)




only one table is required

$E_1E_2R(\underline{A_1}, A_2, A_2, B_1, B_2, B_3, C, D)$

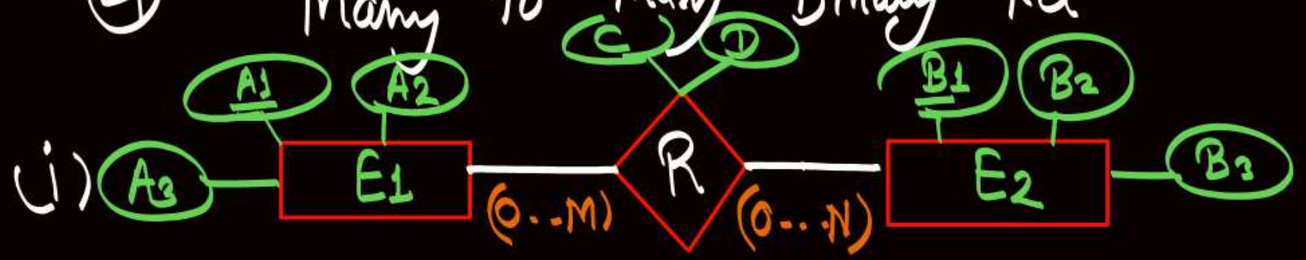
P.K will be from many side rel$^n$



# Topic : ER model to RDBMS table

④ Many-to-Many Binary relation

## (4) Many to Many Binary Relⁿ.



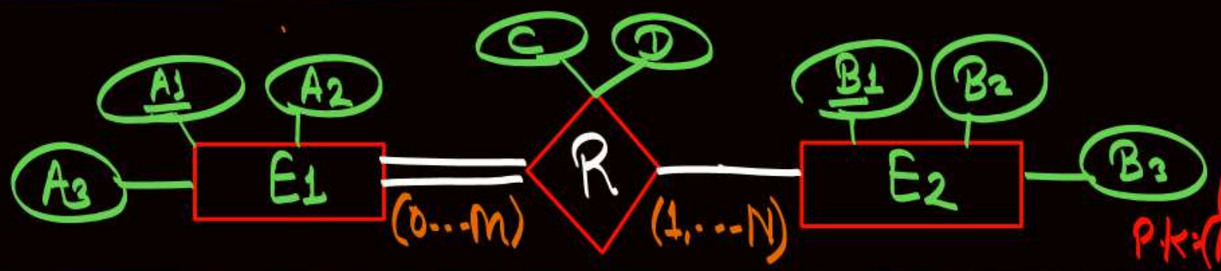

(i)

$E_1(\underline{A_1}, A_2, A_3)$ P.K

$E_2(\underline{B_1}, B_2, B_3)$ P.K

$R(\underline{A_1}, \underline{B_1}, C, D)$ F.K, F.K

---




(ii)

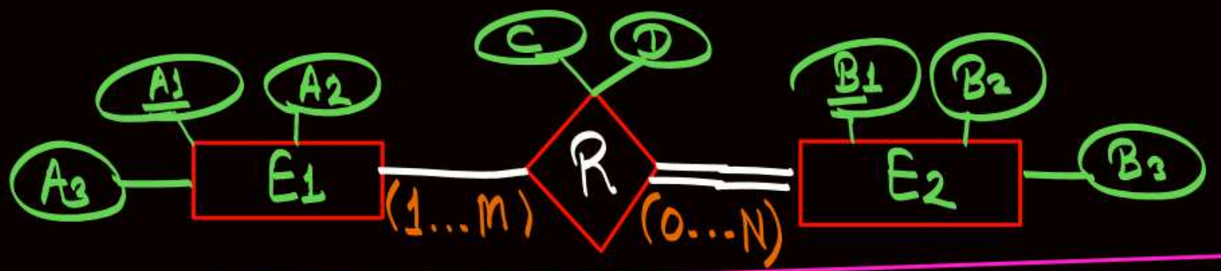
Two tables required

$E_1R(\underline{A_1}, A_2, A_3, \underline{B_1}, C, D)$ FK

$E_2(\underline{B_1}, B_2, B_3)$

P.K = $(A_1, B_1)$

---




(iii)

Same as case (ii)

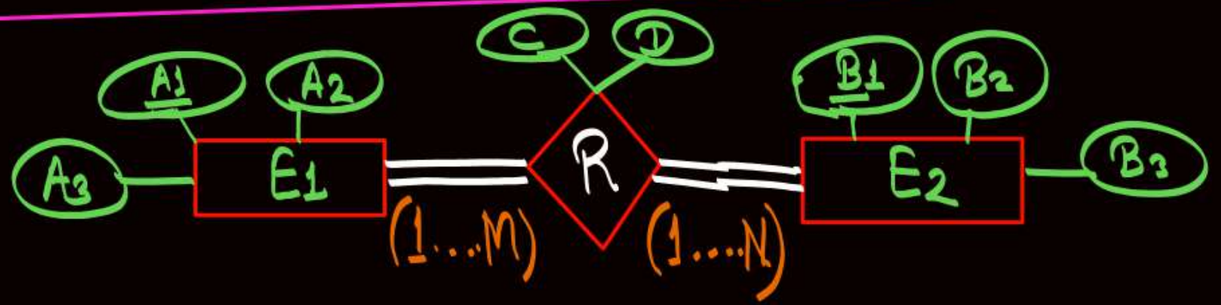
Two tables required

$E_1(\underline{A_1}, A_2, A_3)$

$E_2R(\underline{B_1}, B_2, B_3, \underline{A_1}, C, D)$ FK

P.K = $(B_1, A_1)$

---



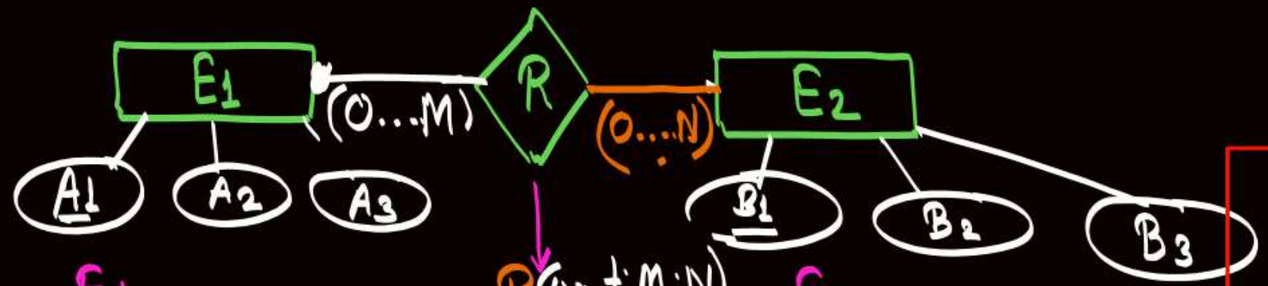

(iv)

only one table is required

$E_1E_2R(\underline{A_1}, A_2, \underline{A_2}, \underline{B_1}, B_2, B_3, C, D)$

P.K = $(A_1, B_1)$

(1)




R(w.r.t M:N)

E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |
| a3 | | |

R

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b3 |

C.k = (A1, B1)

E2

| P.k B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |
| b4 | | |

✓

E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b3 | | |
| a3 | - | - | N | | |
| N | N | N | b1 | - | - |
| N | N | N | b4 | - | - |

Can not be Combined into a Single table, because No Primary can be defined

Primary key can not be defined

E1 R

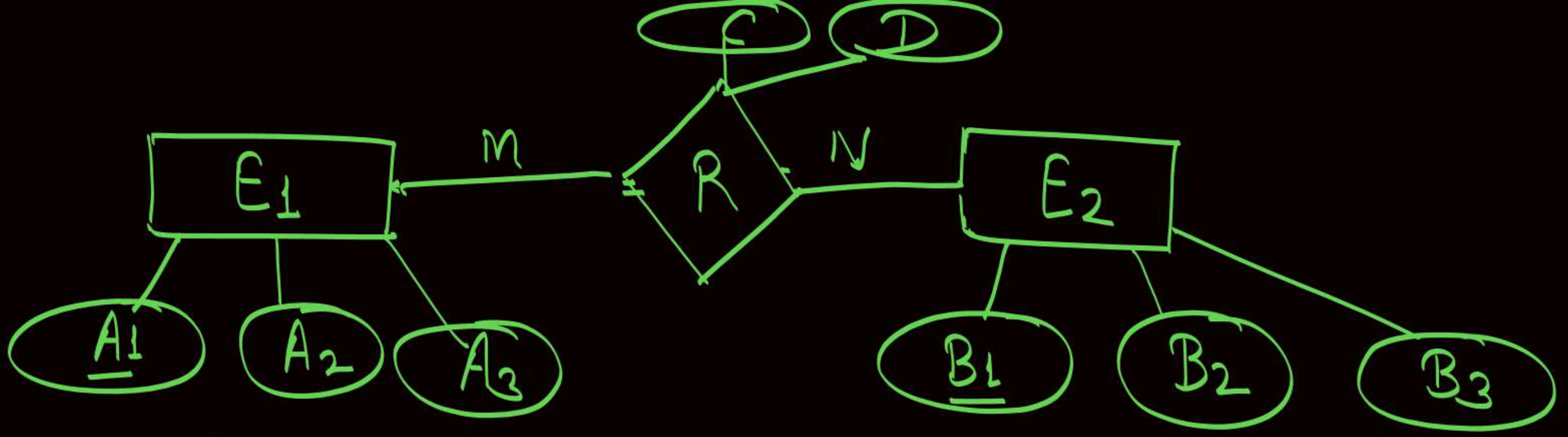
| A1 | A2 | A3 | B1 |
|---|---|---|---|
| a1 | - | - | b2 |
| a1 | - | - | b3 |
| a2 | - | - | b3 |
| a3 | - | - | NULL |

Primary key can not be defined

E2 R

| B1 | B2 | B3 | A1 |
|---|---|---|---|
| b1 | - | - | NULL |
| b2 | - | - | a1 |
| b3 | - | - | a1 |
| b3 | - | - | a2 |
| b4 | - | - | NULL |

(1)




Minimum three tables are required

$E_1(\underline{A_1}, A_2, A_3)$

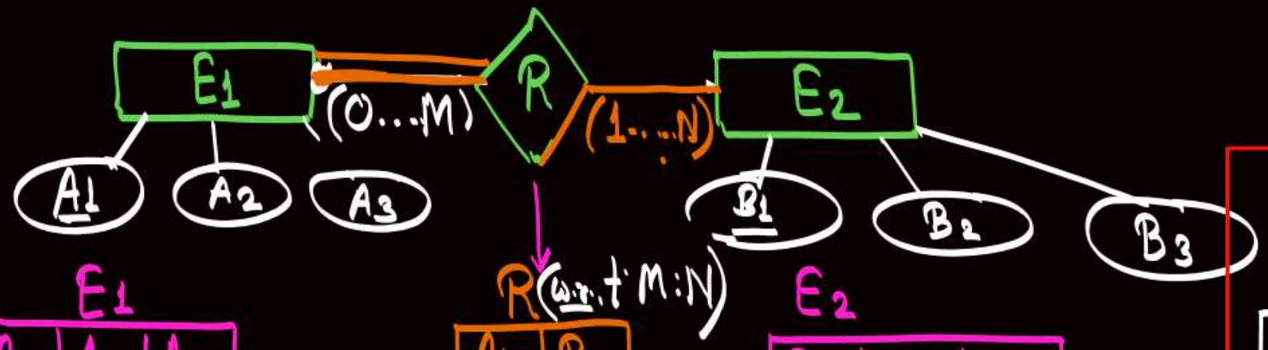
$E_2(\underline{B_1}, B_2, B_3)$

$R(\underline{A_1}, \underline{B_1}, C, D)$

Seperate table will be required for relationship set

{ Primary keys of $E_1$ & $E_2$ together will form primary key of Relation 'R' }

(2)




R (w.r.t M:N)

E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |
| a3 | | |

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b3 |
| a3 | b1 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |
| b4 | | |

✓

E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b3 | - | - |
| a3 | - | - | b1 | - | - |
| IV | IV | IV | b4 | - | - |

Can not be Combined into a Single table, because No Primary can be defined

Primary key will be (A1B1)

E1 R

| A1 | A2 | A3 | B1 |
|---|---|---|---|
| a1 | - | - | b2 |
| a1 | - | - | b3 |
| a2 | - | - | b3 |
| a3 | - | - | b1 |

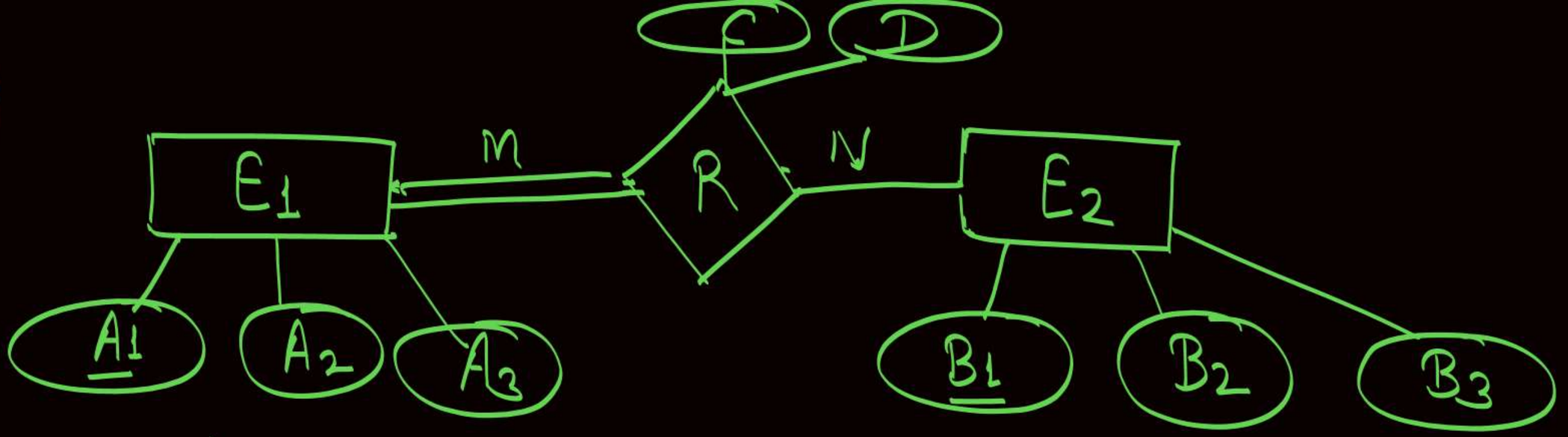
Primary key can not be defined

E2 R

| B1 | B2 | B3 | A1 |
|---|---|---|---|
| b1 | - | - | a3 |
| b2 | - | - | a1 |
| b3 | - | - | a1 |
| b3 | - | - | a2 |
| b4 | - | - | NULL |

→ Relation can not be Combined with entity with Partial Participation

(2)




Minimum two tables are required

F·k

$E_1$ ($\underline{A_1}$, $A_2$, $A_3$, $\underline{B_1}$, $B_2$, C, D)

Primary key

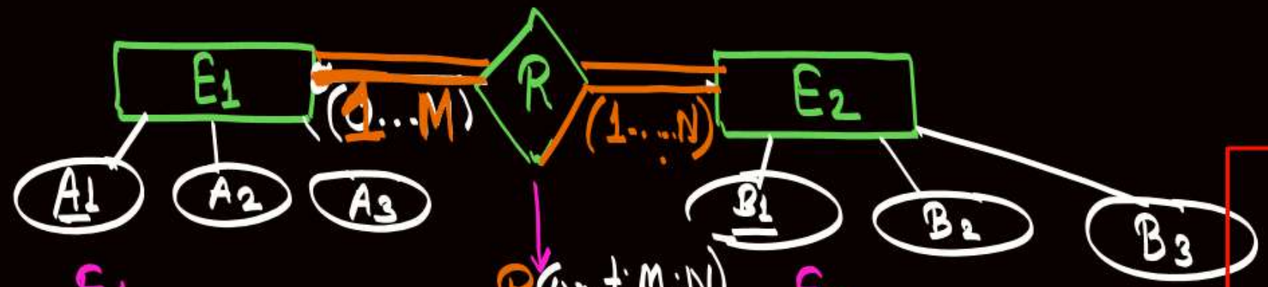
$E_2$ ($\underline{B_1}$, $B_2$, $B_3$)

P·k

{ Primary keys of E1 & E2 together
{ will form primary key of (E1 R)

(3) 3rd case is similar to 2nd case

(4)




E1

| A1 | A2 | A3 |
|---|---|---|
| a1 | | |
| a2 | | |
| a3 | | |

R (wrt M:N)

| A1 | B1 |
|---|---|
| a1 | b2 |
| a1 | b3 |
| a2 | b3 |
| a3 | b1 |

E2

| P.K B1 | B2 | B3 |
|---|---|---|
| b1 | | |
| b2 | | |
| b3 | | |

✓

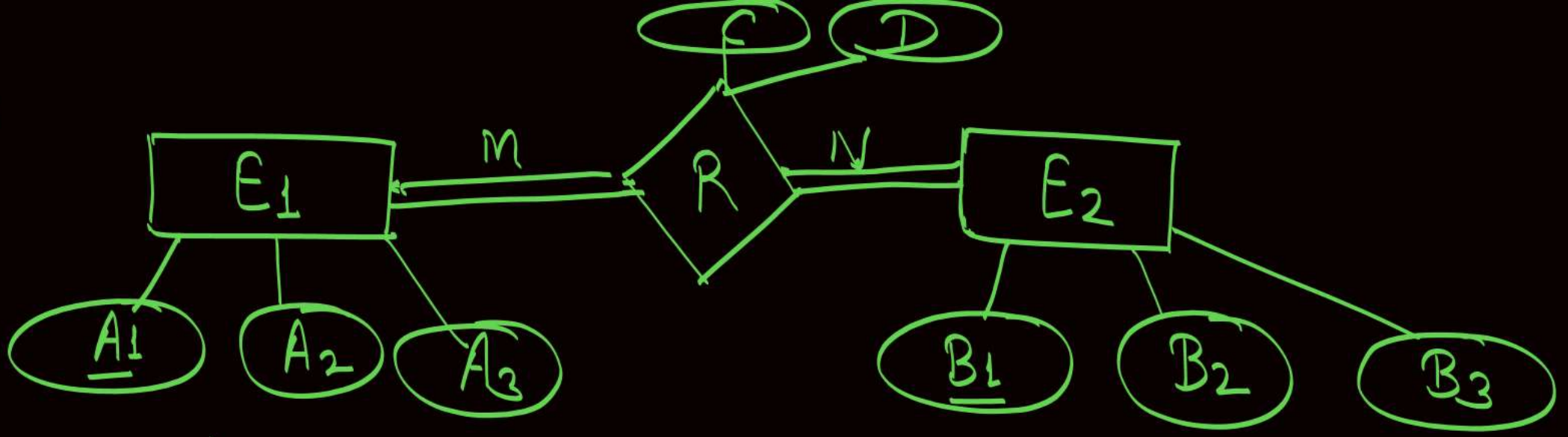
E1 E2 R

| A1 | A2 | A3 | B1 | B2 | B3 |
|---|---|---|---|---|---|
| a1 | - | - | b2 | - | - |
| a1 | - | - | b3 | - | - |
| a2 | - | - | b3 | - | - |
| a3 | - | - | b1 | - | - |

Can be Combined into a Single table,

Primary keys of both the entities together will define primary key of Merged Relⁿ

(4)




only one tables is required

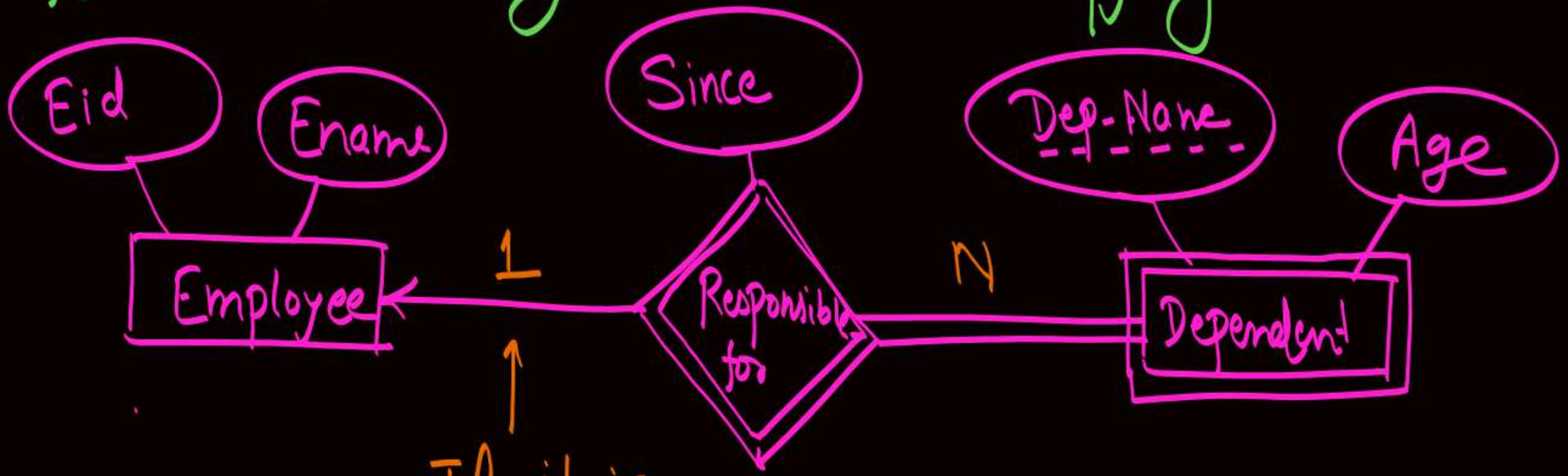
$E_1E_2R(\underline{A_1}, A_2, A_3, \underline{B_1}, B_2, B_3, C, D)$

P.K = $(A_1, B_1)$

{ Primary keys of E1 & E2 together
{ will form primary key of $(E_1E_2R)$

Note:- We will always require a seperate table for weak entity set and identifying relation together

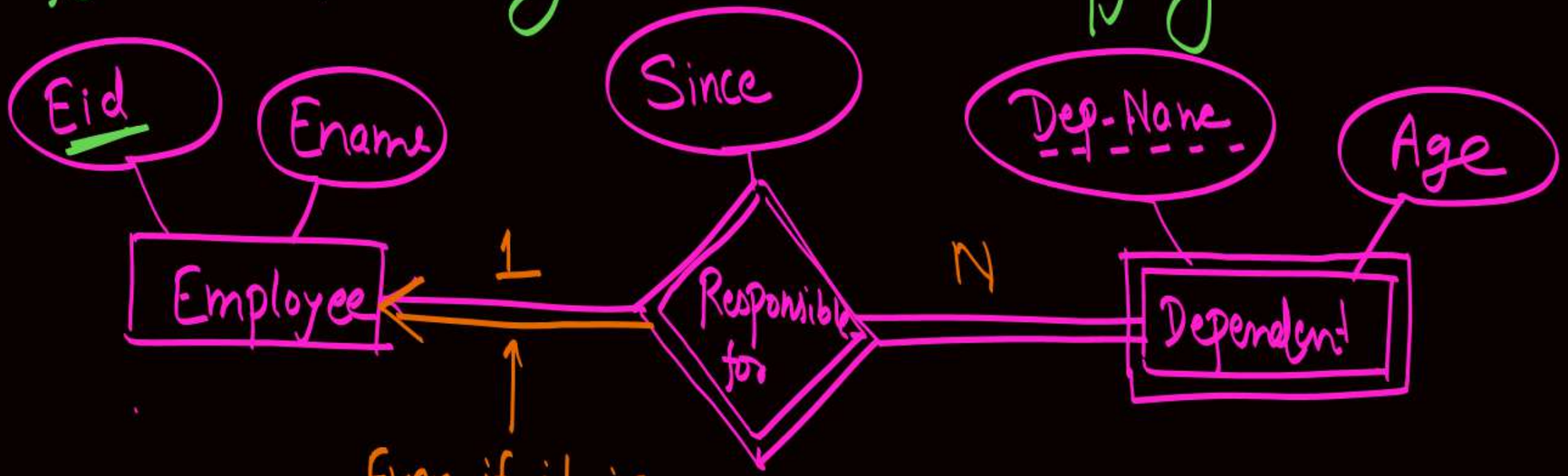
eg.




If it is Partital Participation then any way two tables are required

Note:- We will always require a seperate table for weak entity set and identifying relation together

eg.




Even if, it is a total participation then also two tables will be required bcoz of Weak entity set.

Employee (Eid, Ename)

Responsible_for_dependent (Eid, Dep_name, age, since)

P.k := (Eid, Dep_name)

Note:- W.r.t. multivalued attributes we will always require a seperate table.



# 2 mins Summary

**Topic** ER model to RDBMS table

Slide

# THANK - YOU